

Sultan Of Oman
Ministry of Health
Royal Hospital



سلطنة عمان
وزارة الصحة
المستشفى السلطاني

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The Royal Hospital
Department of Pharmacy

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Title: Management of IV concentrated electrolytes in patient care area

Purpose

To provide a guidance for the restrictions, exceptions, labeling, supply, storage and dispensing of IV concentrated electrolytes in patient care areas, in order to minimize risk and prevent patient harm, as part of the Royal Hospital Management of High Alert Medication strategy (HAM).

Policy Statement

The Royal Hospital will evaluate and limit the availability of concentrated electrolyte products in patient care areas in order to minimize risk and prevent patient harm.

Scope

This policy and procedure applies to all Royal Hospital staff, members of the medical team, including physicians, nurses, pharmacists, pharmacy technicians and other persons working in the Royal Hospital. It also applies when concentrated electrolytes are deemed necessary to be stocked in patient care areas.

Concentrated electrolytes IV injections used for oral or nebulizer use is outside the scope of this policy.

Responsibility

Royal Hospital managers will demonstrate commitment to safety of all patients by ensuring that concentrated electrolytes are not stocked in patient care areas, and if they are stocked in patient care areas (see section Exemptions), that maximal safeguards are in place. The In-patient pharmacy services at the Royal Hospital will demonstrate commitment to safety of all patients by ensuring that processes are in place to identify, store, supply, dispense and label concentrated electrolytes solutions in accordance with this policy.

Principles

The In-patient pharmacy services and ward managers will evaluate and audit bi-annually to limit the availability of concentrated electrolyte solutions in patient care areas, and ensure maximal safeguards are in place to minimize any potential harm caused by concentrated electrolyte solutions.

The following concentrated electrolyte products are restricted by Accreditation Canada Required Organizational Practice and shall not be stored in patient care areas (i.e. ward-stock), unless otherwise specified. (Refer to section. Exemptions)

- Calcium (all salts) in concentrations greater than or equal to 10% (100 mg/mL).
- Magnesium sulfate in concentrations greater than 20% (200mg/mL).
- Potassium (all salts) in concentrations greater than or equal to 2mmol/mL.
- Sodium acetate in concentration greater than or equal to 4 mmol/mL.
- Sodium phosphate in concentrations greater than or equal to 4 mmol/mL.
- Sodium chloride in concentrations greater than 0.9 %.

Exemptions

It may be necessary to have concentrated electrolytes in specific patient care areas. In these cases, an interdisciplinary committee for medication management (Pharmacy and Therapeutics committee) must review, and approve the rational and safeguards that are put in place to minimize the risks of error.

Exempted wards at the time of writing this policy are:

- Critical care areas. (Adult, Pediatrics, & HD)
- Emergency Department
- Operating theatre
- Delivery suite
- Dialysis unit

In those specific patient care areas mentioned, the following must be adhered to:

- Concentrated electrolyte solution must be stored in a locked cabinet with a log book in place and a visible 'high alert medication' label in place.
- Concentrated electrolyte products shall be stored separately from one another and segregated from other products containing potassium or sodium.

Supply of Concentrated Electrolyte Solutions

Patient care areas that do not stock concentrated electrolytes will get a supply from the pharmacy, only when an order is requested from the physician in Al-Shifaa hospital system as a patient specific item.

For specific patient areas that stock concentrated electrolyte solution, the items will be supplied to the specific patient care areas as per the current ward topping up-system in place.

Prior to supplying concentrated electrolyte solutions, in-patient pharmacy services shall label concentrated electrolyte solutions with auxiliary, cautionary, and/or 'high alert medication' labeling as per this policy and the high alert medication policy.

Storage and Labeling

In specific patient care areas that stock concentrated electrolytes, concentrated electrolyte solutions must be stored in an allocated locked cabinet with a log book in place. A visible high alert medication label in the allocated storage cabinet must be present.

Potassium containing solutions must be labeled with further auxiliary and/or cautionary labels as applicable. Concentrated electrolyte should be segregated from other products containing potassium or sodium.

NB: Concentrated electrolyte solutions required for oral or inhalation use must be transferred to a different storage area away from any IV injections and labeled 'For Oral Use Only', or for 'Nebulization Use Only'.

a patient item against an order with all the necessary label requirements.

For exempted inpatient areas, where concentrated electrolytes are stocked, they will be supplied as part of the current inpatient pharmacy topping up system. When concentrated electrolytes are delivered to the ward, they must be segregated from other items and the injections labeled with the appropriate high alert and concentrated electrolyte label to distinguish them from other stock items.

NB If needed, concentrated electrolytes may be available as stand-by stock in certain areas, (after approval). The stock will be replenished against an order as patient supply and replenish the stand-by stock consumed earlier.

Administering

Infusion or syringe pumps will be used to administer infusions of concentrated electrolytes. Smart pumps with error reduction software should be utilized if available.

Independent double check will be completed as per the independent double check policy.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Alberta Health Services Management of High-alert Medications: Concentrated Electrolytes Guideline.	Accreditation Canada QMentum Program, Medication Management Standards	For Surveys Starting After: April, 2016	23

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